

EVIDENCE ROOM

PROFESSIONAL

AP Agent Stack Overview

Evidence Room · AP Agent OS · Pro**Document ID:** ER-APOS-LIB-00**Version:** 1.0 · September 2026**Classification:** Licensed material**Audience:** AP Manager, Controller, Internal Audit, Systems Owner**Companion files:** AGENT_CHARTER_STANDARD.md · RESPONSIBILITY_MODEL.md · A01–A16
charters

1. What this library is

This library is the operating specification for sixteen named AP agents. Each agent has one job, one human owner, a defined starting autonomy level, and a written list of things it must never do.

The stack is ERP-agnostic. It is designed to sit beside SAP S/4HANA, Microsoft Dynamics 365, Oracle Fusion Cloud, NetSuite, Workday, and equivalent payables ledgers. Agents do not replace the ledger. They read from it, prepare work for it, and — only after earned promotion and an explicit control decision — write to it within guardrails.

This is not a chatbot overlay. It is a responsibility model for invoice-to-pay work: who does what, at what autonomy, with what evidence.

Forrester (March 2025) maps current AI use in accounts payable to six clusters: capture, matching, reporting, fraud management, payment management, and e-invoicing/tax. The sixteen agents below cover those clusters and add the work Forrester's list does not name: exception triage, goods-receipt chase, PO quality, supplier and internal follow-up, vendor-statement reconciliation, period close, and root-cause. Fraud-adjacent work is limited to duplicate and anomaly screening. It is not a fraud-detection guarantee.

2. Design rules

1. **One job per agent.** An agent that extracts invoice fields does not release a payment. An agent that flags a duplicate does not conclude fraud. An agent that prepares a close checklist does not sign the close.
2. **Responsibility is earned.** Every agent starts at Level 0 (Observe) or Level 1 (Recommend). Promotion requires measured history, not a vendor demo. See `RESPONSIBILITY_MODEL.md`.
3. **Full autonomy is never the default.** Level 4 is a scoped, time-boxed privilege for a named population of documents. It is not a product setting you switch on at go-live.
4. **Payment authorisation stays human.** The Payment Proposal Review Agent may assemble, score, and challenge a proposal. A human releases the payment file, bank batch, or positive-pay file.
5. **Supplier master changes stay human unless a written policy says otherwise.** The Supplier Resolution Agent drafts communications and change requests. Creation, bank-detail change, and tax-ID change require a named approver. That approval may be configured; the default is human.
6. **No agent is a control.** Agents produce evidence that controls operated. Internal Audit still tests the control. SOX / ICFR ownership remains with the process owner.
7. **ERP is the book of record.** If the agent and the ledger disagree, the ledger wins until a human posts a correcting document.
8. **No guaranteed savings, cycle-time, exception, fraud, compliance, or ROI claims.** Ardent Partners (2024) reports that Best-in-Class AP organisations run at 78% lower cost, 82% faster cycle time, and 59% lower exceptions than their peers, with a 9% exception rate. Those are industry observations, not a forecast for this stack. Measure your own baseline before you claim a result.

3. The sixteen agents

ID	AGENT	PRIMARY JOB	STARTS AT	HUMAN OWNER
A01	Invoice Intake	Capture, classify, extract, and park inbound invoices	1	AP Intake Lead
A02	Invoice Validation	Test completeness, tax, vendor, currency, and coding readiness	1	AP Quality Lead
A03	Matching		1	AP Matching Lead

ID	AGENT	PRIMARY JOB	STARTS AT	HUMAN OWNER
		2-way / 3-way match, tolerances, multi-line allocation		
A04	Exception Triage	Classify, route, and age exceptions	1	AP Exception Lead
A05	Goods Receipt	Detect missing / late / split GRs and chase receivers	0	Warehouse / Receiving Lead
A06	PO Quality	Detect POs that will fail match before the invoice arrives	0	Procurement Operations
A07	Approval	Route, remind, and evidence non-PO and exception approvals	0	AP Manager
A08	Supplier Resolution	Draft supplier queries; propose master-data fixes	1	Vendor Master / AP Communications
A09	Internal Follow-Up	Chase requesters, buyers, and receivers	1	AP Exception Lead
A10	Duplicate & Anomaly	Screen for duplicates and unusual patterns	1	AP Controls Lead
A11	Vendor Statement Reconciliation	Match statements to open items and disputes	0	AP Reconciliations
A12	Payment Proposal Review	Challenge the payment proposal before human release	0	Payments Lead
A13	AP Close	Drive the period-close checklist and accrual completeness	0	Assistant Controller / AP Manager
A14	AP Reporting	Produce the operating pack from ledger facts	1	AP Manager
A15	Root Cause	Cluster repeating exceptions and propose process fixes	0	AP Manager / Process Excellence
A16	AP Manager / Orchestrator	Queue, SLA, hand-off, and autonomy-gate supervision	1	AP Manager

A16 is not a seventeenth processor. It is the dispatcher and the conscience of the stack: it assigns work, enforces autonomy gates, and escalates when an agent is outside its charter.

4. How they interact

4.1 Invoice path (happy path)

Inbound invoice

- A01 Intake capture, classify, extract, create parked document
- A10 Duplicate early duplicate / anomaly screen (blocks or flags)
- A02 Validation completeness, tax, vendor, currency, coding
- A03 Matching 2-way or 3-way against PO / GR
- A07 Approval only if policy requires (non-PO, over-tolerance, new vendor)
- Ledger post human or Level-3 agent under guardrails
- A12 Payment review proposal challenge; human releases

A01 may call A10 before extraction is complete when a header-level duplicate is obvious (same vendor + invoice number + amount). A10 also runs after A02 once lines are reliable.

4.2 Exception path

A02 or A03 or A10 raises exception

- A04 Triage assign reason code, owner, SLA, next action
 - └ GR missing / split → A05 Goods Receipt
 - └ PO incomplete / wrong → A06 PO Quality + buyer
 - └ Supplier data / invoice error → A08 Supplier Resolution
 - └ Requester / receiver silence → A09 Internal Follow-Up
 - └ Duplicate / anomaly hold → A10 (investigation pack) + human
 - └ Approval outstanding → A07

A04 ages the item. A16 escalates when SLA or value gates trip.

A15 reads closed exceptions weekly and proposes causes, not journal entries.

4.3 Periodic path (not invoice-triggered)

CADENCE	AGENTS	OUTPUT
Daily	A16, A14 (flash), A05, A09	Queue health, GR backlog, open chases
Payment run	A12, A10 (payee / bank-detail screen), A07 if needed	Challenged proposal; human release
Weekly	A11 (priority suppliers), A15, A06	Statement gaps, repeating defects, PO quality pack
Month-end	A13, A11, A14, A16	Close checklist, accruals, operating pack
Quarter / promotion review	A16 + process owner + Internal Audit	Autonomy evidence pack

4.4 Data hand-off contract

Agents pass **work objects**, not chat. A work object is a structured record with:

- object_id, object_type (invoice, exception, GR chase, statement line, payment proposal line, close task)
- source_system + source_doc_id
- company_code / subsidiary
- vendor_id, amount, currency, due_date
- reason_code (from the shared taxonomy)
- autonomy_level_in_force
- recommended_action + confidence + evidence_refs[]
- human_owner + sla_due
- state (new / in_review / waiting_external / ready_to_post / posted / rejected / escalated)

A16 is the only agent allowed to reassign human_owner or change state to escalated. Downstream agents may propose a state change; they do not silently close another agent's object.

4.5 Shared exception taxonomy

Use these codes everywhere. Do not invent local synonyms.

CODE	MEANING	TYPICAL NEXT AGENT
HDR-ERR	Header incomplete or unreadable	A01, A08
LINE-ERR	Line, UoM, or tax-line extraction failure	A01, A08
VND-UNK	Vendor not found or ambiguous	A08
VND-BLK	Vendor blocked, one-time, or payment hold	A08, A12
TAX-ERR	Tax code, jurisdiction, or e-invoice tax mismatch	A02
CUR-ERR	Currency / FX / company-code mismatch	A02
P0-MISS	No PO, invalid PO, or PO closed	A06, A09
P0-QLTY	PO exists but cannot match (account, vendor, price, qty, date)	A06
GR-MISS	Expected GR not posted	A05
GR-SPLIT	Partial / multi-GR allocation needed	A05, A03

CODE	MEANING	TYPICAL NEXT AGENT
QTY-VAR	Quantity outside tolerance	A03, A09
PRC-VAR	Price outside tolerance	A03, A09
DUP-SUS	Duplicate suspected	A10
ANOM-SUS	Pattern anomaly, not a known duplicate	A10
APPR-PEND	Approval outstanding	A07
STMT-UNM	Statement line unmatched	A11
ACCR-GAP	Received-not-invoiced or invoiced-not-received gap	A13
PAY-HOLD	Payment-proposal challenge	A12
CTRL-BRK	Control or SOD concern	A16 + Controls Lead

5. What stays human

The following remain human at every autonomy level. An agent may prepare the file. It may not complete the act.

DECISION	WHY IT STAYS HUMAN	AGENT THAT PREPARES
Payment file / bank batch / positive-pay release	Cash leaves the company	A12
Approval of invoices above the written matrix	Delegation of authority is a human control	A07
Vendor create; bank-detail or tax-ID change	Classic fraud path; configurable but default human	A08
Write-off, tolerance override above policy, or account reclass above limit	P&L / balance-sheet judgement	A03, A04, A13
Fraud conclusion, SAR/internal-investigation referral	Legal and HR process	A10
Period-close sign-off	Controller accountability	A13
Autonomy promotion or demotion	Governance, not a model score	A16 + owner + Audit

DECISION	WHY IT STAYS HUMAN	AGENT THAT PREPARES
Prompt / rule / tolerance table change	Change control	Process owner
Legal, tax-position, or statutory interpretation	Licensed advice is out of scope	A02 may flag; Tax owns
SOD override (same person create + approve + pay)	Control failure	A16 blocks; Audit reviews
If a local policy wants an agent to do any of the above, that is a Level-4 request and requires the promotion pack in RESPONSIBILITY_MODEL.md plus Internal Audit acknowledgement. The Evidence Room default is: do not .		

6. ERP-agnostic operating notes

6.1 Two books, one stack

Many groups run more than one ledger. The worked example in every charter — **Northline Industrial Group** — runs SAP S/4HANA for the core companies and NetSuite for one acquired entity. The agents share a taxonomy and a work-object schema. They do **not** share a single posting API.

Implementation rule: one **adapter per ledger**. The charter stays the same. The adapter translates:

STACK CONCEPT	SAP S/4HANA	D365 F&O / BC	ORACLE FUSION	NETSUITE	WORKDAY FINANCIALS
Parked / draft invoice	MIR7 / parked MIRO / FB60 hold	Pending vendor invoice	Incomplete Payables invoice	Vendor bill (pending approval)	Supplier invoice (draft)
Posted invoice	RBKP / BSEG	Posted vendor invoice	Payables invoice	Vendor bill	Supplier invoice
Purchase order	EKKO / EKPO	Purch table	PO header / line	Purchase order	Purchase order
Goods receipt	MIGO / MKPF / MSEG	Product receipt	Receiving	Item receipt	Receipt
2-/3-way match	MIRO match + OMR6 tolerances	Invoice matching policy	Matching rules	Vendor bill matching	Receipt / invoice match

STACK CONCEPT	SAP S/4HANA	D365 F&O / BC	ORACLE FUSION	NETSUITE	WORKDAY FINANCIALS
Vendor	Business Partner / LFA1	Vendor	Supplier	Vendor	Supplier
Payment proposal	F110 proposal	Payment journal	Payment process request	Payment batch	Settlement run
Statement	Vendor line items + FBL1N	Vendor aged / statement	Payables trial	Vendor statement	Supplier statement
Close	MMRV / GR/IR / F.13	Period close workspace	Close monitor	Period close	Close

6.2 Identity and keys

Do not assume a global vendor number. Northline's SAP vendor 4002187 (Helion Fasteners NA) is NetSuite vendor v-1184 at the acquired entity. The adapter stores a **cross-walk**. Agents key work objects on (source_system, source_vendor_id) and display the group alias.

Invoice numbers are not unique across vendors. The natural key is (company_code, vendor_id, invoice_number, invoice_date, gross_amount, currency) plus any legal e-invoice UUID where one exists.

6.3 Comma lists, attachments, and e-invoices

- Header PDF / XML / UBL / PEPPOL / country e-invoice formats land in A01. Do not force a single capture engine. The charter requires an extract, a confidence, and a human-review path — not a named OCR brand.
- Attachment stores (OpenText, VIM, ReadSoft, native ERP attach) are tools, not agents.
- Where a statutory e-invoice identifier exists, A01 persists it. A02 tests it. A10 uses it as a hard duplicate key.

6.4 What the adapter must never do

- Post in one ledger because a twin document posted in another.
- Recycle a payment proposal across systems.
- Treat a NetSuite “pending approval” bill as a SAP-posted invoice.
- Write back to the ledger without the autonomy gate and the named posting user (system user with logged human authoriser).

7. Operating model for a 14-person AP team

The Northline worked example is illustrative (Evidence Room framework). It is the same company used in every agent charter so the library reads as one system.

Northline Industrial Group — 4,200 employees; about 18,000 invoices per month; SAP S/4HANA plus one NetSuite entity (Northline Pacific Components); 14-person shared-services AP team.

Suggested role map (illustrative, not a staffing recommendation):

ROLE	HEADCOUNT (ILLUSTRATIVE)	OWNS AGENTS
AP Manager	1	A16, A14, A15, A07
Intake / capture	2	A01
Validation / matching	3	A02, A03
Exception specialists	3	A04, A09
Vendor master & supplier comms	2	A08
Payments & bank	1	A12
Reconciliations & close	1	A11, A13
Controls / quality	1	A10, A02 (second)
Warehouse liaison (not AP headcount)	—	A05
Procurement operations (not AP headcount)	—	A06

A16's queue should make this split visible. If matching staff are working supplier emails, the orchestrator is mis-routed — that is a control and a productivity issue.

8. Autonomy at go-live (default)

LEVEL	MEANING	AGENTS AT GO-LIVE
0 Observe	Read, log, report. No recommendations that look like instructions to the ledger.	A05, A06, A07, A11, A12, A13, A15
1 Recommend	Propose an action. Human accepts, edits, or rejects.	A01, A02, A03, A04, A08, A09, A10, A14, A16
2 Prepare		None at go-live

LEVEL	MEANING	AGENTS AT GO-LIVE
	Draft the document, email, or parking entry. Human posts or sends.	
3 Execute within guardrails	Post / send inside written limits.	None at go-live
4 Managed autonomy	Operate a named population with exception-only review.	Never default

Promotion criteria, demotion triggers, and the evidence pack live in `RESPONSIBILITY_MODEL.md`. A16 refuses to dispatch a Level-3 action if the agent's current authorised level is lower.

9. Control spine

Every agent charter repeats a control block. The stack-level spine is:

1. **Written policy** — tolerance table, approval matrix, vendor-change policy, payment-release policy. Versioned. Dated. Owner named.
2. **Autonomy register** — which agent, which company code, which document population, which level, review date. Owned by AP Manager; visible to Internal Audit.
3. **Immutable work log** — every recommendation, override, post, and escalation. Retention aligned to financial-record policy (set locally; do not assume a number).
4. **SOD** — the system user that posts is not the user that releases cash. Agents inherit this. A16 blocks a path that would collapse maker and checker.
5. **Change control** — prompts, match rules, reason codes, and tolerance numbers change through the same path as an ERP configuration change.
6. **Completeness** — inbound invoice count (mailroom / e-invoice / portal / EDI) reconciled to A01 created objects, daily.
7. **Evidence pack** — each agent lists what Internal Audit can pull without a screen-share.

Ardent Partners (2024) Best-in-Class exception rate is 9%. Use that as an external benchmark when you present your own exception rate. Do not treat 9% as a target the agents are warranted to hit.

10. Measurement spine

Four KPI families. No vanity metrics (messages sent, “AI interactions”, unanchored “accuracy %”).

FAMILY	QUESTION	TYPICAL STACK MEASURES
Activity	Did the agent touch the work?	Objects ingested, recommendations issued, coverage of inbound volume
Operational	Did the process move?	Cycle time receipt-to-post, first-pass match, exception aging, chase response
Financial	Did money move correctly?	Duplicate \$ prevented (confirmed, not suspected), discount captured vs offered, GR/IR balance, on-time pay
Risk	Did controls hold?	Override rate, false-positive investigation load, SOD blocks, unauthorised-level attempts, late close tasks

Each agent charter defines formula, source, cadence, and owner. A14 publishes the pack. A16 uses the pack to recommend promotion or demotion — it does not promote itself.

11. Implementation sequence (Evidence Room framework)

Do not stand up sixteen agents on day one. Sequence is a control decision.

WAVE	STAND UP	WHY THIS ORDER
0	A16 + autonomy register + work-object schema + inbound completeness	Without a dispatcher and a log, you cannot evidence anything
1	A01, A02, A10	You cannot match or triage garbage extract; you must screen duplicates early
2	A03, A04	Matching and triage are the volume engines
3	A05, A09, A08	Most exceptions die in GR, requester silence, or supplier data
4	A06, A07	Upstream PO quality and approval routing once exception codes are stable
5	A12, A11, A13	Payment, statements, close — after the daily path is trusted

WAVE STAND UP

WHY THIS ORDER

6 A14, A15

Reporting and root-cause require a quarter of clean objects

Wave dates are local. The order is not.

12. Failure modes of the stack (not of a single agent)

FAILURE	WHAT GOES WRONG	WHO OWNS THE FIX
Adapter drift	Field mapping silently drops tax or PO line	Systems Owner + A02 completeness test
Taxonomy fork	Teams invent local reason codes; A15 becomes noise	A16 + AP Manager
Autonomy creep	A processor “just lets it post” without a register update	Controls Lead; demote
Completeness gap	Portal invoices bypass A01; late invoices appear at close	A01 + A13
Cross-ledger double pay	Same supplier invoice in SAP and NetSuite	A10 + A12 + cross-walk
Chat instead of objects	Work happens in email; audit trail dies	A16 rejects non-object hand-offs
Vendor demo promotion	Level 3 granted because extraction “looked good”	Promotion pack refused

13. Cost monitoring (stack)

Monitor agent cost the same way you monitor a BPO invoice: unit cost, exception cost, and rework.

COST OBJECT	WHAT TO CAPTURE	OWNER
Capture / model / API	Cost per invoice ingested (A01) and per exception classified (A04)	Systems Owner
ERP API volume	Reads vs writes; write spikes are a control event	Systems Owner
Human minutes	Review, override, chase, investigation	AP Manager

COST OBJECT	WHAT TO CAPTURE	OWNER
False-positive load	Hours spent clearing A10 / A03 noise	Controls Lead
Rework	Reversal / credit / re-post after agent-prepared documents	Assistant Controller
Idle licence	Agents switched on with no objects	A16

A14 reports these monthly. Cost reduction is a measurement, not a claim.

14. What “implementation-ready” means here

A charter is implementation-ready when a project team can:

1. Name the human owner and the backup.
2. Map every required input to an ERP field or a file.
3. Configure the reason codes and the routing table.
4. Produce the listed artifacts on day one at Level 0 or 1.
5. Show Internal Audit the evidence list.
6. Refuse any action in the exclusions list.
7. Score the agent after 60–90 days using the KPI block and decide stay / promote / demote.

If a section does not tell you what to do, how, who owns it, what can go wrong, how you control it, how you measure it, or what evidence proves it, the section is incomplete. Fix the charter before you configure the agent.

15. Document control

ITEM	VALUE
Owner	Evidence Room — AP Agent OS product
Change path	Version increment; changelog; no silent edits to autonomy defaults
Related product folders	../Governance · ../Controls · ../KPI_and_Measurement · ../Process_Mapping · ../Testing
Worked example	Northline Industrial Group (fictional). Never replace with a real employer or client.

ITEM	VALUE
Advice status	Not legal, tax, accounting, or investment advice. Independently authored.

16. File index

FILE	CONTENTS
00_AGENT_STACK_OVERVIEW.md	This document
RESPONSIBILITY_MODEL.md	Levels 0–4; promotion; demotion; evidence pack
AGENT_CHARTER_STANDARD.md	Required fields for every agent
A01_invoice_intake.md	Invoice Intake Agent
A02_invoice_validation.md	Invoice Validation Agent
A03_matching.md	Matching Agent
A04_exception_triage.md	Exception Triage Agent
A05_goods_receipt.md	Goods Receipt Agent
A06_po_quality.md	PO Quality Agent
A07_approval.md	Approval Agent
A08_supplier_resolution.md	Supplier Resolution Agent
A09_internal_follow_up.md	Internal Follow-Up Agent
A10_duplicate_anomaly.md	Duplicate & Anomaly Agent
A11_vendor_statement_reconciliation.md	Vendor Statement Reconciliation Agent
A12_payment_proposal_review.md	Payment Proposal Review Agent
A13_ap_close.md	AP Close Agent
A14_ap_reporting.md	AP Reporting Agent
A15_root_cause.md	Root Cause Agent
A16_ap_manager_orchestrator.md	AP Manager / Orchestrator

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